

Trent Wiles

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EDUCATION

Northeastern University — Khoury College of Computer Sciences

Boston, MA

Candidate for Bachelor of Science in Computer Science and Criminal Justice

August 2023 — Present

- GPA: 3.7/4.0; Dean's List for all semesters; expected graduation: May 2027
- Clubs & Organizations: ACM Chapter, Content Club, OSCRR Student Board Member
- Relevant Coursework: Object-Oriented Programming, Database Design, Software Development, Algorithms & Data Structures, Introduction to Math Reasoning, Discrete Math, Networking Fundamentals

EXPERIENCE

Amazon

June 2026 - August 2026

Loss Prevention Intern

- Will contribute to an intern project related to loss prevention at an Amazon Fulfillment Center

HITE Hedge Asset Management LLC

January 2026 - June 2026

Software Engineer Co-op

- Built a React and TypeScript AI chatbot UI under senior supervision that gives portfolio managers natural language access to commodities and automotive data from internal S3 and Databricks sources, surfacing company metrics to support investment decisions
- Deployed the chatbot to the company's production web infrastructure on EC2 and hardened it with Microsoft OAuth and firewall-level IP whitelisting to block unauthorized external access and maintain regulatory compliance
- Eliminated the repetitive cycle of downloading data, editing in Excel, and re-feeding it to the model by embedding a live-syncing CSV editor with formula support directly into the chat interface

Kaleidoscope at Khoury College

September 2025 - Present

Co-President

- Will co-lead a student organization of 10 members alongside a fellow Co-President, coordinating events and initiatives with a faculty advisor to support Khoury College's club ecosystem
- Led the software team to manage and investigate technical resources for Northeastern Kaleidoscope, a club council overseeing 50+ student organizations funded by Khoury College
- Partnered with the operations team to automate Slack notifications integrated with Notion, streamlining communication across 10,000+ dollars in annual club distributions

PROJECTS

HackerNews Clone | *Go, React, TypeScript, PostgreSQL, Docker*

July 2025

- Built a production-grade HackerNews clone with magic link authentication, submission/comment posting with reCAPTCHA v3 protection, and community moderation flagging, deployed via Docker and Caddy
- Implemented full voting and sorting systems, user customization features, and comprehensive HTTP/business logic logging to enable 2,000+ monthly active users
- Engineered a Go backend and React/TypeScript frontend using ShadCN components, achieving sub-100ms API response times and support for 500+ users in production

RAG Pipeline Metrics | *Python, gRPC, PostgreSQL, Clickhouse, Cassandra, OpenSearch*

June 2025

- Partnered with NetApp Instaclustr in a team of 8 during Northeastern's Software Engineering in Australia program to design a RAG pipeline using Mistral 7B that benchmarked query latency across four distributed database backends
- Developed scalable FastAPI microservices for document ingestion and embedding generation, leveraging gRPC for modular inter-service communication in a fast-paced Agile environment
- Optimized document processing pipeline using SPECTER2 embeddings to achieve 10,000 documents per hour throughput, demonstrating production-grade scalability for scientific PDF ingestion

Flag | *PHP, MySQL, Apache, Cloudflare*

December 2020

- Built a video sharing platform enabling users to upload, stream, and rate videos with a MySQL database backend and PHP server-side rendering, achieving 240+ hosted videos and 100 concurrent users
- Self-hosted the application on Ubuntu Linux using Apache HTTP Server on a home network while leveraging Cloudflare for video caching and content delivery, demonstrating early DevOps and infrastructure management skills

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS, Go, C, Kotlin

Frameworks: Node.js, Flask, ExpressJS, React, WSGI, JUnit, Jest, JQuery

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Redis

Tools: Git, GitHub, Bash, Linux, Notion, LaTeX, Docker, DigitalOcean, Caddy, Apache2, Nginx

INTERESTS

Cycling, Traveling, Urban Exploration, Tennis, Kayaking